Solving systems of equations with a variety of methods

1 Describe the graphical solution of a system of two linear equations.

2 Find the number that should be multiplied to the second equation of the following system of equations to eliminate y using addition method:

$$\begin{cases} -3x - 12y = 6 \\ -2x - 4y = -4 \end{cases}$$

3 If the first equation is multiplied by 3, find the number that should be multiplied to the second equation of the following system of equations to eliminate y using addition method:

$$\begin{cases} 4x + 4y = 6 \\ 5x + 3y = 3 \end{cases}$$

4 Consider the equations $y = 3x - 1$ and $y = -5x + 23$.

a Complete the given table by finding the y-values for each of the given x-values:

b Hence, find the values for x and y which satisfy both the equations.

x	$y = 3x - 1$	$y = -5x + 23$
1		
2		
3		
4		
5		

5 Consider the equations $y = 5x$ and $y = 9 - 4x$.

a Complete the given table by finding the y-values for each of the given x-values:

b Hence, find the values for x and y which satisfy both the equations.

x	$y = 5x$	$y = 9 - 4x$
1		
3		
5		

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6 Consider the following linear equations:

$$y = \frac{x}{3} + \frac{1}{3} \text{ and } -8y = 8x + 8$$

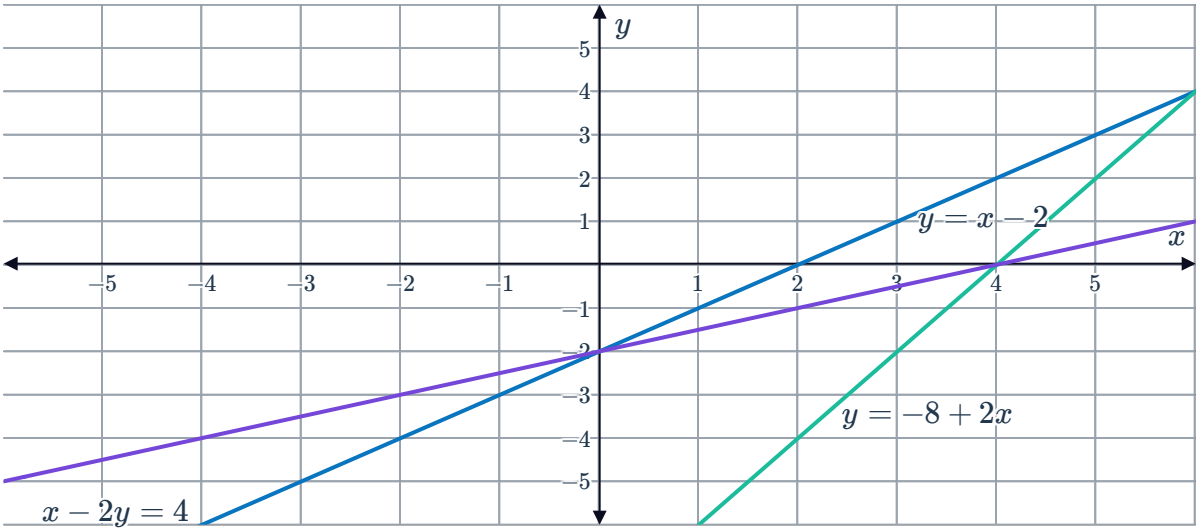
a Find the intercepts of $y = \frac{x}{3} + \frac{1}{3}$.

b Find the intercepts of $-8y = 8x + 8$.

c Sketch the lines of the two equations on the same coordinate plane.

d Hence, state the values of x and y which satisfy both equations.

7 Use the given graph to solve each system of equations:



a $y = x - 2$
$y = -8 + 2x$

b $y = x - 2$
$x - 2y = 4$

c $y = -8 + 2x$
$x - 2y = 4$

8 Solve the following systems of equations:

a $2x - 5y = 1$
$-3x - 5y = -39$

b $2x - 5y = -10$
$y = -5x + 6$

c $-\frac{x}{4} + \frac{y}{5} = 8$
$\frac{x}{5} + \frac{y}{3} = 1$

d $5x + 3y = 15$
$\frac{6y}{5} = 10 - 4x$

e $-7p + 2q = -\frac{13}{10}$
$-21p + 10q = -\frac{9}{10}$

f $-5p - 7q = -\frac{43}{5}$
$-15p - 28q = -\frac{157}{5}$

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9 Consider the following linear equations:

$$-6x - 2y = -28$$
$$2x + 16y = 40$$
$$4x - 2y = 12$$

a Find the values of x and y that satisfy the first two equations.

b Determine if this solution satisfies the third equation by substituting the values of x and y into the left hand side of the equation.

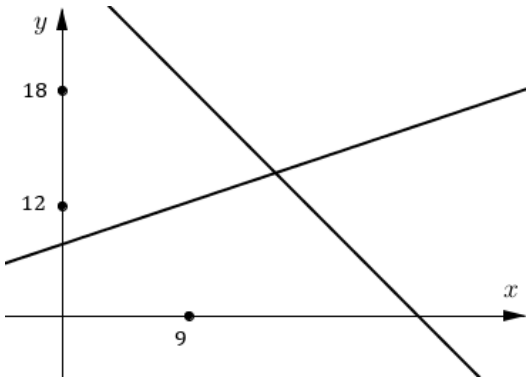
c Hence, state whether the lines are concurrent.

10 Suppose two numbers add up to 9 and have a difference of 1. Let x represent the smaller number and let y represent the other number. Write a system of equations that describes this problem.

11 Marge graphed a system of two linear equations. She began with an approximation of $x = 9$ and it did not satisfy both equations simultaneously. One equation gave a resulting value of $y = 12$ and the other equation gave a resulting value of $y = 18$ as shown. Note that the graph is not to scale.

a State the closer approximation to the solution: $x = 5$ or $x = 13$.

b She substituted $x = 13$ into both equations and found that it did not satisfy the two equations simultaneously. She also found that the difference between the resulting y values was 10.
State the x value closest to the solution: $x = 9$ or $x = 13$.



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12 Consider the system of linear equations.

$$\text{Equation 1: } y = x - 6$$
$$\text{Equation 2: } y = 5x + 6$$

a The following system of equations was obtained by adding a multiple of Equation 2 to Equation 1.

$$\text{Equation 3: } y = x - 6$$
$$\text{Equation 4: } 5y = 21x + 18$$

Find the multiple of Equation 2 that was added.

b Find the values of x and y that satisfy Equation 3 and Equation 4.

c Is Equation 4 a multiple of Equation 1?

13 Consider the system of linear equations.

$$\text{Equation 1: } y = x + 8$$
$$\text{Equation 2: } y = -7x + 16$$

a Find the values of x and y that satisfy both equations.

b Add Equations 1 and 2 to create Equation 3. State Equation 3 in the form $y = mx + b$

c Multiply Equation 1 by the number 3 to create Equation 4. State Equation 4.

d Find the values of x and y that satisfy Equations 3 and 4.

e Are the solutions in the part (d) the same as for the system of Equations 1 and 2? Explain your reasoning.

14 Consider the equation $\frac{3x}{2} - 5 = \frac{2x}{3}$.

a Solve for the value of x that satisfies the equation.

b To verify the solution graphically, which two equations would need to be graphed?

- Equation 1: $y = \frac{2x}{3}$
- Equation 2: $y = 4$
- Equation 3: $y = 9x - 30$
- Equation 4: $y = \frac{3x}{2} - 5$

c Graph the equations in part (b) on the same plane.

d Hence, find the values of x that satisfies the two equations simultaneously.

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15 Glen is looking for a mobile phone plan to sign up to. TeleNow has a plan that charges an initial fee of $55c$ per call, plus $3c$ per minute of calls. Konnex has a plan that charges an initial fee of $30c$ per call, plus $8c$ per minute of calls.
Let x be the duration of a call in minutes and y is the total cost for each phone call in cents.

a Complete table of values for each plan.

i TeleNow's plan

ii Konnex's plan

x	1	3	5	7	9
y					

b State whether each equation is the correct cost equation for TeleNow and Konnex's plans.

i TeleNow: $y + x = 55 + 3$, Konnex: $y + x = 30 + 8$

ii TeleNow: $y = 55 + x$, Konnex: $y = 30 + x$

iii TeleNow: $y = 3x + 55$, Konnex: $y = 8x + 30$

iv TeleNow: $y = 55x + 3$, Konnex: $y = 30x + 8$

c Graph the equations for TeleNow and Konnex's plans on the same plane.

d Find the duration of a phone call that would cost Glen the same on either plan.

e If Glen usually makes phone calls of 6 minutes or longer, which plan should Glen choose?

16 Consider the system of linear equations.

$$\begin{aligned} x + 2y &= -1 \\ -4x + 4y &= 1 \end{aligned}$$

State whether each system of linear equations has the same solution as the one given.

a $\begin{aligned} 3y &= -12 \\ x + 2y &= -1 \end{aligned}$

b $\begin{aligned} 12y &= -3 \\ x + 2y &= -1 \end{aligned}$

c $\begin{aligned} 2x &= -4 \\ x + 2y &= -1 \end{aligned}$

d $\begin{aligned} 2x &= -4 \\ -4x + 4y &= 1 \end{aligned}$

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Additional questions

17 Solve each system of equations:

a

$$\begin{cases} 5x + y &= 4 \\ 45x + 4y &= 16 \end{cases}$$

b

$$\begin{cases} y &= x + 2 \\ x + 2y &= 16 \end{cases}$$

c

$$\begin{cases} 7x + y = 65 \\ 3x + y = 33 \end{cases}$$

d

$$\begin{cases} x &= 5 - 2y \\ 3x + 4y &= 7 \end{cases}$$

e

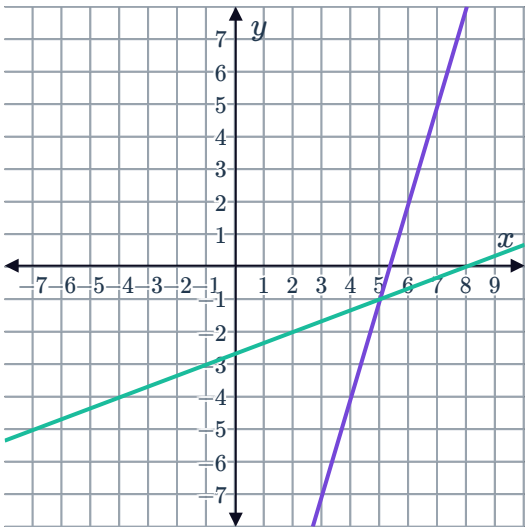
$$\begin{cases} x + 4y &= 15 \\ -x + 3y &= -1 \end{cases}$$

f

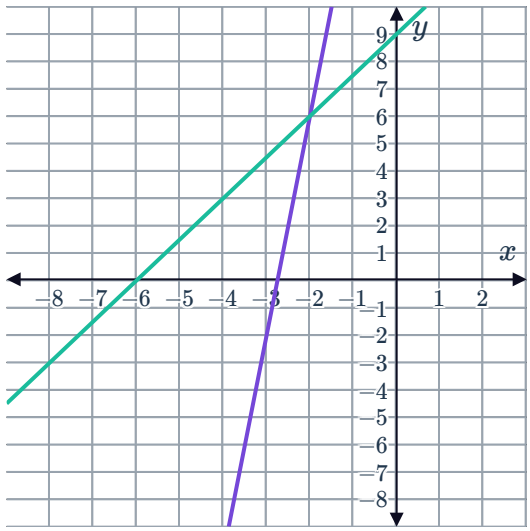
$$\begin{cases} 5x - 7y &= 3 \\ -3x - 7y &= -69 \end{cases}$$

18 Solve each system of equations for their point of intersection:

a

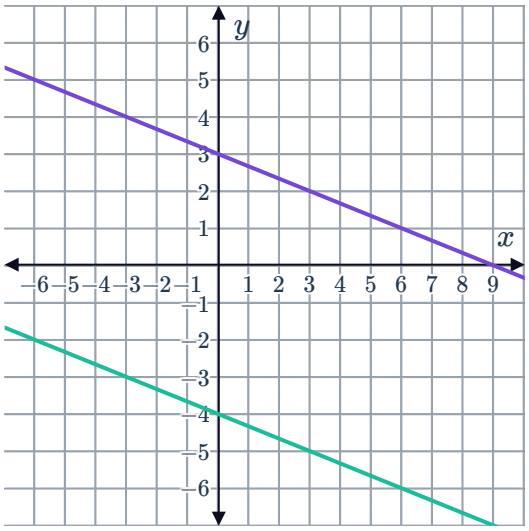


b

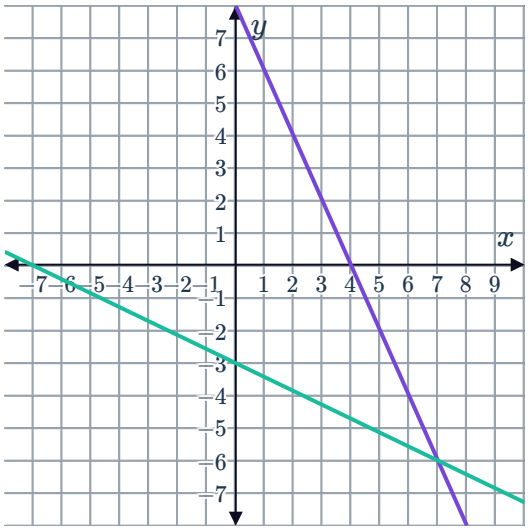


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c



d



19 Solve each system of equations:

a

$$\begin{cases} y = 4.5x + 22 \\ 2.5x + 3y = 16 \end{cases}$$

b

$$\begin{cases} -6x - 38y = 2 \\ -2x - 19y = 34 \end{cases}$$

c

$$\begin{cases} y = 101 - 15x \\ y = -3x + 49 \end{cases}$$

d

$$\begin{cases} x = 75y - 400 \\ x = 38y + 53 \end{cases}$$

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